

Practicing equity in the AI ecosystem: Co-designing solutions for sociotechnical challenges

Monique Munarini*

Computer Science

University of Pisa

Pisa, Italy

monique.munarini@phd.unipi.it

Abstract

My research aims to assess the performance of Artificial intelligence (AI) audits using an equity approach. I explore equity as a guiding value in the deployment of accountability mechanisms to assess compliance with the EU AI Act (AIA). I use feminist theory to conceptualise equity as a translational value between different fields in the AI ecosystem that can foster reliable AI. The research integrates legal, ethical, and technical considerations to develop and validate a framework that ensures the reliability of AI audits while protecting fundamental rights. The methodology includes legal analysis, feminist research, and empirical investigation through co-design and participatory workshops. The research timeline spans three years, with a focus on building a concept of equity, developing indicators for the framework, and collecting inputs from workshops. By leveraging equity as a guiding value, my research aims to develop actionable frameworks useful to the community involved in the AI development and deployment.

CCS Concepts

• **Human-centered computing** → *Empirical studies in collaborative and social computing*; • **Applied computing** → Law, social and behavioral sciences; • **Computing methodologies** → **Artificial intelligence**.

Keywords

reliable AI, equity, co-design, accountability

ACM Reference Format:

Monique Munarini. 2025. Practicing equity in the AI ecosystem: Co-designing solutions for sociotechnical challenges. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3706599.3707636>

1 Background and motivation

The transformative potential of AI systems across various domains is well recognized. These systems are sociotechnical artefacts, integrating not only technical components like data and algorithms but

also social factors, including the inputs of designers and the applications by deployers [17]. When considering AI's role in advancing the United Nations 2030 Agenda for Sustainable Development, Vinuesa et al.[45] highlighted a paradox: while AI could contribute to achieving some of the Sustainable Development Goals (SDGs), it may also exacerbate challenges like gender inequality and global disparities. The unequal distribution of AI's benefits and harms stems largely from biased systems that disproportionately impact marginalised groups. Notable examples include women disqualified from job applications by biased hiring algorithms at Amazon [16], wrongful accusations of fraud leading to tragic outcomes in welfare systems [29], and AI chatbots like Microsoft Tay [40], which have propagated hate speech. Beyond these headline cases, there are numerous overlooked harms, such as the misidentification of trans individuals in airports, resulting in humiliating experiences [14] or people who do not have a chance to apply for a job because the algorithm did not show it to them due to bias in the optimization feature [2]. In response to these disparities, a wave of non-binding ethical guidelines emerged at various levels of governance from international organisations, nations to private companies [21, 28], followed by binding regulations aimed at mitigating the risks of AI [12]. The European Union AI Act (AIA), effective as of August 2024 [43], represents a key step toward regulating AI through a risk-based approach. However, there is growing concern about the overreliance on third-party accountability mechanisms, raising questions about the effectiveness of such audits [15, 35]. One of the accountability mechanisms in the AIA is the Fundamental Rights Impact Assessment (FRIA) to be conducted by deployers of high-risk systems [43]. Recent research reveals that none of the FRIA studied implemented sufficient methodological control, leaving gaps in transparency and accountability [6]. Given the sociotechnical nature of AI systems, a purely technical approach to governance and auditing falls short.

Building on the work of the Human-Computer Interaction (HCI) and FAccT (Fairness, Accountability, and Transparency) communities on fairness and algorithmic fairness [44], it is evident that operationalising fairness remains challenging due to its contextual nature and the limitations of debiasing mechanisms [20, 32, 47, 48]. These challenges have led to a shift away from the original goal of using fairness to uphold fundamental rights such as equality and non-discrimination in AI system design and deployment. I propose equity as an alternative to fairness, especially given the lack of consensus on what constitutes [38]. Drawing from feminist theory, I have developed a definition of equity, which underpins a co-designed, participatory research framework. Feminist theory and design are rooted in the goal of including marginalised voices

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3707636>

empowering them through meaningful participation in the design, deployment and monitoring processes [5, 24]. This framework, rooted in design justice principles [14], will inform the development of indicators for assessing AI auditing methodologies with a focus on upholding fundamental rights through an equity lens.

2 Research objectives

This research aims to build a framework that will be used to improve accountability of AI systems from a sociotechnical perspective focusing on the avoidance of inequitable outcomes to marginalised groups. First, it sets forth a working definition of equity from a feminist perspective as setting common terminology is essential to efficient multi-disciplinary collaboration [25]. I seek to demonstrate how the proposed definition of equity positions this principle in the intersection of law, policy-making, and technological developments related to AI Governance, and how it can be used as an alternative to fairness to address issues involving bias. Second, using the design justice principles [14] I built indicators to operationalise equity focusing on integrating social and technical aspects of the AI ecosystem [41].

- To provide a common definition of equity that can be used and understood by many stakeholders within the AI lifecycle
- To create an equity-based framework that can be used by different actors in the AI ecosystem to take informed decisions in selecting methodologies aligned with the ethical value equity.
- To use an equity-based framework to assess current auditing methodologies for AI models.
- To incorporate equity-by design [10] in auditing methodologies, so ethical considerations such as equity can be systematically assessed throughout the AI lifecycle, leading to more reliable and accountable AI systems.

3 Methodology and approach

An intuitive survey of scholarly knowledge was conducted to find the ways in which equity is addressed in different fields of knowledge in the AI ecosystem and where the operationalisation of this value could be more impactful in upholding fundamental rights. The literature review involved an analysis of ethical guidelines from different levels of governance, such as the UNESCO AI principles, OECD AI Principles, EU Ethics Guidelines for Trustworthy AI [37]. The EU AI Act (AIA) is a regulation with a global impact due to its potential extraterritorial feature, also known as Brussels' effect [27]. For this reason, the project focuses on this legislation. Nevertheless, the review also explored EU legislation on fundamental rights, such as the EU Charter of Fundamental Rights, EU Non-Discrimination regulation and EU Gender Mainstreaming Resolution [36]. From the technical perspective, it assesses projects aiming equitable outcomes such as the *Queer in AI* [39] project and the EU Horizon project *Fairness and Intersectional Non Discrimination in Human Recommendation (FINDHR)* [13] that has the goal to develop non-discriminatory algorithms used in recruitment processes. The second year of the PhD was mostly focused on refining the standard definition of equity and developing the equity-based framework with its indicators. The use case chosen for this research are AI systems to support decision-making process

in the recruitment context. This research uses an intersectionality-informed approach based on Design Justice Principles [14] to design the social-technical indicators to operationalise equity. To conclude the second year, inspired in the co-design and participatory action research were deployed to design the workshop to validate the equity indicators with experts from the above mentioned FINDHR Project. The third year will be devoted to develop the second workshop with the participation of marginalised groups and to write about the findings.

4 Results and contribution to date

My dissertation comprises a three-part research investigation aiming to address the research objectives in Section 2. By the time of CHI2025 part 1 (defining equity) and 2 (develop and validate framework) will be successfully concluded. Part 1 was recently presented at ICEGOV 2024 (1-4 October 2024) [37] and awarded the best research paper of the conference. Part 2 will have the results submitted to Facet 2025 in January 2025. The third and final part is proposed in Subsection 5.2. The feedback from CHI 2025 Doctoral Consortium on the first and second parts will be essential to shape the third one.

4.1 Defining equity

The motivation for focusing on equity arises from the limitations of fairness, particularly in addressing systemic inequalities as highlighted in Section 1. Feminist theory emphasizes recognizing and addressing power imbalances in society [22, 46]. Haraway [23] calls for this ability to translate knowledge between fields and power-differentiated communities to avoid reductionist solutions when she defines translation as an enabler of power-sensitive conversations. Defining equity was inspired by the articulation of new paradigms on HCI research [18, 26] redefining the way in which sociotechnical dimensions are analysed considering the intrinsic multidisciplinary nature of emerging technologies such as AI systems.

The objective, therefore, is to operationalise an ethical value that has a translational function within the AI ecosystem and global governance to mitigate the escalation of unbalanced outcomes affecting marginalised groups, while embracing the sociotechnical and multidisciplinary nature of AI systems. The results were shared in the paper I recently presented at ICEGOV 2024 and received the award of best research paper [37]. Equity is not about treating everyone the same or differently according to a most often protected category but about providing those in need, regardless of its association in a specific group, with the necessary resources to achieve a sense of belonging within their communities. This focus on belonging [33] is central to equity, as it ensures that marginalised peoples are not merely included superficially but are empowered to participate meaningfully in decision-making processes [31]. The differences between equity and fairness are represented in Table 1. In this research equity is defined as **"providing the necessary resources to those who need them to foster a sense of belonging in a specific community"** [37]. There are four main elements to be considered: (i) identifying those in need, (ii) determining the essential resources, (iii) understanding the target community, and (iv) achieving a sense of belonging within that community. To

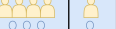
	WHO APPLIES?	HOW ARE THEY SELECTED?	WHO IS HIRED?	FINAL OUTCOME
EQUITY				
FAIRNESS				

Figure 1: A comparison between Equity and Fairness in a hiring process. The aim is to highlight how Equity emphasises providing equitable outcomes, while Fairness emphasises providing a fair process.[37]

better understand these elements, a speculative recruitment scenario using fairness and equity approaches to seek a more diverse workforce was designed in Fig1. From an equity approach, those in need are the job applicants who faced or still face any barrier to compete on the same ground with other job applicants due to societal issues. The target community is the workforce that the hired candidate will join. The achievement of belonging is rooted in the meaningful inclusion [34] of the new member removing barriers and power-imbalances with its new peers. The resources needed to make this feasible might include internal policies tackling systemic inequalities considering their employees. A fairness approach will be guided towards ensuring a fair process in the hiring process where some candidates from specific categories might receive different treatment if they identify as a legally protected group. This is a key aspect of the leaking pipeline that contributes to the gender gap in tech [42], for example. Although positive actions may recruit women, measures to support their promotion and reduce the burden of caregiver responsibilities are often lacking. The process to hire them may be fair, but the outcomes do not uphold equality and non-discrimination. Women might even face additional discrimination, as colleagues might perceive them as hired through a different process.

4.2 Operationalising equity

In the scope of operationalising the equity definition, using design justice framework [14], I developed a framework based on key indicators as points of concern regarding equity. This involves collaborative practices to address sociotechnical challenges that may arise from AI deployment, also present on feminist design work [5]. The indicators were designed to be deployed with a holistic perspective [36], going beyond the AI system to the whole net of interactions between humans and algorithms. Seven sociotechnical indicators were designed to be tested in the workshops. In the case of an AI system used to screen CVs more than just the lifecycle of the system (design, training, deployment, monitoring phases), the indicators should be checked against the pipeline in which the system is embedded, like processes involving the ones who operate the system (headhunters), who audit the system (third-party auditors), who will be impacted by the outcomes (recruitment companies, candidates, hiring companies). As these stakeholders change over time, this is not a one time protocol, but a constant process to verify if equity is ensured within this AI ecosystem.

5 Next steps

There are two milestones that will mark the following steps on this research:

5.1 Co-design workshop

The workshop titled 'How to think about equity in the AI field?' comprises two dimensions focusing on co-design and design justice principles [14]. The first one is to test and refine the equity indicators with practitioners from multiple fields working with AI systems in recruitment. The second is to include the knowledge from diverse voices that might face unbalanced outcomes from AI systems. The idea of a workshop as method came from Participatory action research with the awareness of its limitations [9]. I use constructive design process [30] to understand the different structures that each dimension might require. Within HCI, Dombrowski et al. [19] proposed social-justice strategies when working with societal challenges like the ones raised in this research.

The Horizon project FINDHR (Fairness and Intersectional Non Discrimination in Human Recommendation) [13] is an interdisciplinary project aiming at detecting, preventing and avoiding discrimination in AI systems used in recruitment processes. The consortium works on capacity-building initiatives, new technologies to measure discrimination risks and technical guidance to measure impact assessments of AI systems. The experts involved in FINDHR are key stakeholders on addressing discrimination and the development of AI-based systems. As one of my supervisors is working on this project, I was able to connect with the principal investigators and suggest a collaboration. The workshop was advertised to the FINDHR team with the following summary: The workshop aims to empower participants with the tools and understanding needed to navigate ethical considerations in AI development and deployment, emphasising the importance of diverse perspectives and inclusive practices in shaping the future of AI. It explores the intersection of equity and AI, emphasising the need to protect fundamental rights in AI technologies used to support decision-making processes in recruitment. Participants will engage in interactive activities designed to deepen their understanding of equity and practical indicators for integrating an equity-based approach into AI systems. The workshop will begin with an engaging icebreaker, ensuring they understand the focus of the day and key definitions related to equity and fairness in AI. The main session involves a co-design activity centred on a case study of a recruitment company facing challenges with AI-based hiring processes. Participants will work through indicators such as intersectionality, access to information, contestability, and debiasing mechanisms to devise sociotechnical recommendations for promoting diversity and inclusion within the AI lifecycle. Through this interactive workshop, participants will gain insights into operationalising equity within AI systems, focusing on acquiring actionable insights and tools to promote equity in their respective AI projects, contributing to the advancement of reliable and inclusive AI technologies". The workshop was developed in the FINDHR annual meeting on 13 November 2024 in a 1h30 session. The outcomes of the workshop are under analysis using thematic analysis [8]. The second workshop will follow the same structure of the workshop with the FINDHR team and it is aimed to be deployed between March-April 2025.

Table 1: A summary comparison of equity and fairness [37]

Value	Scope	Focal point	Approach	Definition
Equity	Outcome-focused	Collective	Aiming an equitable society dictates how individuals will be treated	Understanding and providing to the ones who need the necessary resources to belong in a particular community
Fairness	Process-oriented	Individual	Tailored process that will define how individuals will be treated and this will shape a fair society	<i>Ethics</i>: “Treat equals equally and unequals unequally” [4] <i>Law</i>: A process or decision is considered fair if it does not discriminate against people based on their membership to a protected group (EU perspective) [3] <i>Computer Science</i>: Typically treated in a reductionist way, seeing fairness only as a quantifiable statistical metric [11]

Some Civil Society Organisations and research labs were contacted as potential partners with positive feedback.

5.2 Assessing accountability mechanisms

The last objective of my PhD research involves assessing accountability mechanisms that are used for compliance with provisions from the EU AI Act, mostly the Fundamental Rights Impact Assessment on article 27 [43]. I will limit my study to publicly-available mechanisms as the aim of this assessment is to meaningfully contribute to HCI and AI fields and only publicly available documentation can inform decisions for potential use of these tools by the technical community.

The evaluation criteria can be divided on a scale from 1 (gold standard) to 5 (substandard). The tiers are defined in terms of (i) how much the accountability mechanisms are aligned with the specified evaluation criteria, in other words, satisfy the proposed equity indicators (ii) how much room for revision they leave, and (iii) to what extent the demonstrated level of effort is commensurate with the stakes.

The results will be displayed in a matrix in a user-friendly way. In AI Governance the use of such matrix became very popular after the ones regarding the assessment of LLM models according to the EU AI Act [7] or AI safety frameworks [1]. It intends to provide a visual guide to stakeholders, including the technical community, to understand how much aligned or not a given accountability mechanism when choosing a compliance tool. The CHI2025 Consortium will also help me to understand to what extent this is also relevant to HCI and AI practitioners.

5.3 Benefits statement and challenges

One of the open questions that I aim to address at CHI2025 is to understand to what extent my framework is something that reaches the community working in the design and development of AI systems, or what can be adapted to achieve this goal. As this is a multidisciplinary research involving legal analysis, ethical and feminist theorists, governance frameworks and technical assessments, the development of the methodology and methods involving co-design and participatory research increase the inherent challenge of serving on the best way possible all the fields involved in it, also included when seeking feedback as it is the case of this Doctoral Consortium.

As this PhD research uses the ethical value equity to bridge AI governance initiatives from the legal and policy-making side with technical developments, one of the main challenges that does not depend on my research is the actual implementation of the EU AI Act and the standard process that are still in development to bring effectiveness to this piece of legislation.

6 Dissertation status and long-term goals

I am currently a second-year PhD candidate at the Italian National PhD in AI for Society in the Computer Science Department at the University of Pisa advised by, in alphabetical order, Dr James Brusseau (Pace University NY/University of Trento), Dr Lorenzo Angeli (University of Trento), and Dr Salvatore Ruggieri (University of Pisa). My final goal with this PhD research is to present a framework that bridges compliance requirements with policy and technological developments aimed at upholding fundamental rights such as equality and non-discrimination. Considering my legal (BA in Laws at Unicuritiba/Brazil) and political science (MA in Human Rights and Multi-level Governance at University of Padova/Italy) background, my research is multidisciplinary which poses advantages and limitations. Actively joining the CHI community is a strategy to overcome some limitations as the community seems very inclusive of these types of research. The completion date of my Ph.D. studies is expected to be Spring 2026. Upon graduation, I aim to continue working on multidisciplinary research focused on equitable AI systems.

Acknowledgments

Work partly supported by the European Union’s Horizon Europe research and innovation programme for the project FINDHR (g.a. No. 101070212). Views and opinions expressed are those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

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